The Red/Black Game III:
Belief Carrying, Safe Re-Solving, and the Certificate Problem

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Abstract

Paper II solved the abstraction G' of heads-up Red/Black to stated per-class tolerances (its 2026-08 colour-swap audit later bounded how much of G' that pins down; see the remark in Section 2) and certified that the solved profile is exploitable in the real game G for at least 0.598 per game of a 2.0 range. This paper pursues the other direction: an agent together with a computed upper bound on its real-game exploitability. There are two results and one honest accounting. First, carrying beliefs closes the measured gap. Re-solving each round of play under the exact public posterior, tracked by a filter over the factorized belief family of Paper I and validated to 10^{-12} against brute-force enumeration, produces a dose-response in the depth of carrying: the strongest probe's lower bound falls from 0.598 with no carrying to 0.078 [0.016, 0.139] with full carrying, a 7.7× collapse. The belief reset was most, but demonstrably not all, of the measured exploitability: the full-carry interval excludes zero. Second, because naive re-solving admits adversarial counterexamples and certifies nothing, we build the machinery a guarantee requires: a safe-resolving gadget whose per-resolve margins are measured live by exact best-response sweeps, and a composition theorem bounding the agent's exploitability by a root residual plus the worst reachable chain of margins, with no opponent model anywhere in the bound. Third, the accounting: assembling today's measured margins yields only the trivial upper bound. An earlier assembly that claimed 1.958 is dissected as a cautionary result; it substituted an on-path average for a supremum the opponent controls. The obstruction is localized in a single quantity, the honorability of the leaf-price surface at layers $T' \geq 4$, and a four-item program targets it, each item motivated by a measurement reported here. The sandwich read [0.078, 2] at this paper's first close; as of 2026-08 the deployed store of record reads $\varepsilon_G \geq -0.005$ $[-0.040, +0.030]$ at $n = 3{,}200$, a probe reading of zero, so the sandwich for the deployed agent is $[-0.005, 2]$ with the upper jaw still trivial (the dated remarks of Section 8). Every number is regenerated by scripts accompanying the papers.

1 Introduction

Papers I and II left heads-up Red/Black in a precisely known position. The real game G cannot be tabulated (2.4×10^{33} infosets). Its reachable beliefs live in a rank-one family (Paper I, Thm. 10.3). The abstraction G', which resets beliefs at every round boundary, is tabulable, and Paper II solved it. Discarding the carried beliefs costs at least 0.598 per game of a 2.0 payoff range in real-game exploitability, a bound certified by Paper II's local best-response probe, which wins actual games against the solved profile.

That last number is a lower bound, and no lower bound can answer the project's remaining question:

Exhibit an agent for heads-up Red/Black together with a proof that no opponent strategy beats it by more than ε_G, for a small, computed ε_G.

A probe that fails to find an attack has measured its own reach; only an argument that quantifies over all opponent strategies yields an upper bound.

The sandwich. Every claim in this paper lives between two jaws: probe lower bounds (Paper II's local best-response family, extended here with resolver-aware variants whose opponent model is byte-exact) and the certificate upper bound. The jaws must never cross; a certificate below a valid probe reading refutes the certificate rather than the probe. We report both jaws in every results table, and the honest summary of this paper is that the lower jaw sat at 0.078 at its first close and reads zero for the deployed store of record (-0.005 [-0.040, $+0.030$] at $n = 3{,}200$; Remark 8.4), while the upper jaw sits at the trivial cap of 2. A second discipline governs the upper jaw: a bound that assumes the opponent plays in any particular way, including the way our own stored profile plays, guarantees nothing, so every certified quantity below is either computed offline over all opponent types or measured live by an exact best-response sweep that includes the opponent's certified alternative.

The target. The project's bar is $\varepsilon_G \leq 0.10$ per game (5% of the value range), with 0.05 as a stretch goal. Alongside it we require sandwich consistency, no defensive regression against the unsafe prototype, and head-to-head and bot-battery parity under Paper II's standing battery.

Contributions. First, a dose-response finding (Section 4): re-solving under carried beliefs drives the strongest probe's lower bound from 0.598 with no carrying to 0.078 [0.016, 0.139] with full carrying, a 7.7× collapse whose interval excludes zero. Second, safe-resolving machinery (Sections 5–6): a gadget whose per-resolve margins are measured live by exact best-response sweeps, and a composition theorem bounding exploitability by a root residual plus the worst reachable chain of margins, with no opponent model in the bound. Third, an honest accounting (Section 8): today's certified upper bound is the trivial one, an earlier claim of 1.958 is dissected as a cautionary result, and the remaining distance is localized in the honorability of the leaf-price surface at layers $T' \geq 4$, which the four-item program of Section 9 targets.

2 Preliminaries

This section collects every object the rest of the paper uses.

The games. G is heads-up Red/Black, a two-player zero-sum game of hidden information; all values are seat-0 payoffs in $[-1, 1]$, so the payoff range is $U = 2$. G' is the belief-reset abstraction of G (Paper I, Definition 10.6): the game in which, at every round boundary, both players' beliefs are replaced by the prior of the boundary's class, discarding the tilts accumulated during play. Paper II solved G' class by class; its solution (the *stored profile*, with *stored values* at every class) is the raw material for everything below.

Remark 2.1 (What the stored values determine, 2026-08). Paper II's colour-swap audit (`scripts/check_symmetry` showed that the stored second-generation profile determines the values of G' only to about 0.03 and its type-conditional values to about 0.37: CFR+ pins each class's aggregate value but selects arbitrarily among per-type-pair decompositions, and the bootstrap composes the selection noise. Every leaf price this paper's resolves read from the stored surface carries that indeterminacy, which is of the same order as the per-resolve leaf dishonorability measured in Section 8.

How much of the measured leaf term is intrinsic to depth-limited resolving and how much was the stored surface being undetermined is answered by re-running these instruments against the third-generation unique-selection solve (Paper II, Section on the colour-swap audit).

Boundaries, classes, layers. A *boundary* b is a full public history ending at a procurement resolution. Boundaries group into the canonical *classes* of Paper I; $\kappa(b)$ denotes the class of b, λ_κ its prior over type pairs, and K_b its deal-coupling kernel. Classes are stratified into *layers* indexed by T', which decreases by one per round of play: boundaries occur at $T' = 9$ down through $T' = 2$, so a game contains at most $L_{\max} = 8$ resolved boundaries (re-derived by script).

Beliefs. At a public history h ending at a boundary of class κ, the exact joint posterior over type pairs factorizes as $\mu_h \propto \lambda_\kappa \cdot g_0 \cdot g_1$, where g_0, g_1 are nonnegative per-seat *tilt* vectors over the seats' types (Paper I, Thm. 10.3); the reachable belief domain is exactly this rank-one family. The deal couples the seats only through the invariant $a = r + x$ of each type (Paper I, Lemma 10.2), which is what makes the factorization survive play. g_b is the agent's exact own tilt at boundary b.

Meters. For a strategy pair $\sigma = (\sigma_0, \sigma_1)$ in G, with u_p seat p's expected payoff, the real-game exploitability is

$$\varepsilon_G(\sigma) \;:=\; \max_\tau u_1(\sigma_0, \tau) \;+\; \max_\tau u_0(\tau, \sigma_1),$$

the total best-response gain over both seats: zero at equilibrium, never above $U = 2$ (the trivial cap). For two distributions μ, λ over type pairs,

$$\mathrm{TV}(\mu, \lambda) \;=\; \tfrac{1}{2} \sum_{(s,t)} \bigl|\mu(s,t) - \lambda(s,t)\bigr|;$$

a single reported TV number is the expectation of $\mathrm{TV}(\mu_h, \lambda_{\kappa(h)})$ over the boundaries h visited in play, weighted by reach (Paper II's TV measurement, 2,000 games).

Probes. A *local best-response* (LBR) probe is an opponent that, at each of its decisions, best-responds with a bounded lookahead against the agent's actual continuation strategy, to which it has exact access. Its realized winnings over played games certify a statistical lower bound on ε_G (mirrored seeds, Wilson 95% intervals): the probe wins real games, so no accounting error can inflate the bound. This is Paper II's probe family; the resolver-aware variants introduced here additionally hold a deterministic shadow copy of the resolver, checked byte-exact, so that probing a deterministic re-solving agent uses an opponent model that is the agent.

Counterfactual values. For opponent type t at a reachable boundary b, $\mathrm{CBV}_A(b, t)$ denotes the supremum, over opponent play below b, of the opponent's true conditional counterfactual value in G against the agent's continuation, computed under the $K_b(\cdot, t)\, g_b$ counterfactual weighting. The composition theorem of Section 6 is an inductive bound on CBV_A.

3 The belief filter

The agent's belief tracking applies Paper I's structure theorems directly; its correctness burden is discharged once, here, for everything downstream.

The filter. At a boundary of class κ the posterior is $\mu_h \propto \lambda_\kappa \cdot g_0 \cdot g_1$ (Section 2), and the filter tracks the two tilt vectors by per-round transfer sums: each seat's factor is updated by the likelihood of the public record under that seat's strategy, with the deal coupling riding along unchanged ($a = r + x$ is invariant). Against an independent brute-force replay of all consistent histories, the filter reproduces the exact posterior to 10^{-12}. The 2,000-game TV measurement of Paper II cost 64 seconds *including* play, so exact belief tracking is computationally free at this game's scale.

Deviation conditioning. A re-solving agent deviates from the stored profile, so its *own* transfer must use its actually played strategies (the filter accepts a per-seat solution override); the opponent's transfer uses the stored profile as a model. A shadow-replay check, run for both the unsafe and the safe resolver (Sections 4 and 5), pins this down: replaying a full game's public record through the filter reproduces the resolver's tables bit for bit.

Conditionals that survive any opponent. The opponent-side transfer is a model, and a guarantee cannot lean on it. It does not have to.

Lemma 3.1 (Model-free conditionals)**.** *For any opponent strategy τ and public history h, the conditional distribution of the agent's type given the opponent's type is* $\Pr(s \mid t, h) \propto K^{\text{deal}}(a(s), a(t))\, L_A(s)$*, where L_A is the agent's own likelihood sum. It does not depend on τ, and the filter update preserves it under any strictly positive opponent weighting, because a-invariance makes the deal kernel constant along both seats' transitions.*

Proof. The deal couples the seats only through (a_0, a_1), deterministic invariants of the current types. Path masses factorize per seat given the coupling, and the opponent's factor multiplies whole columns, cancelling from every conditional. Every own transition preserves $a(s)$ and every opponent transition preserves $a(t)$, so the kernel factor pulls out of both transfer sums unchanged. $\square$

As a check, support-only opponent transfers reproduce the exact filter's conditionals to 10^{-10} on played traces (`test_support_filter_conditionals`).

Two measured warnings sharpen the lemma's value. The tilts are not the seats' standalone likelihood products: the layer prior is a different kernel at every class, and naive per-seat products miss the kernel drift by TV up to 0.129 on played traces. And a zero-likelihood opponent trace (an opponent the stored model thinks impossible) would break a naive filter; the safe agent instead swaps the opponent transfer for its structural support indicator and keeps resolving, which by Lemma 3.1 preserves every conditional the certificate consumes. Robustness against arbitrary opponents has to be engineered in exactly this way. In 2026-08 the support-indicator transfer was promoted from fallback to a primary deployment mode and ran 3,200 probed games with zero breaks and zero fallbacks (artifacts `probe_solve-5-pruned-osf_a.json` and `_b.json`), validating it as a break-proof mode for stores whose support is narrower than their opponents' play.

4 Re-solving without a guarantee

The prototype changes exactly one input of Paper II's machinery. At a boundary with public history h, it solves the entry class's one-round subgame with entry weights μ_h in place of the

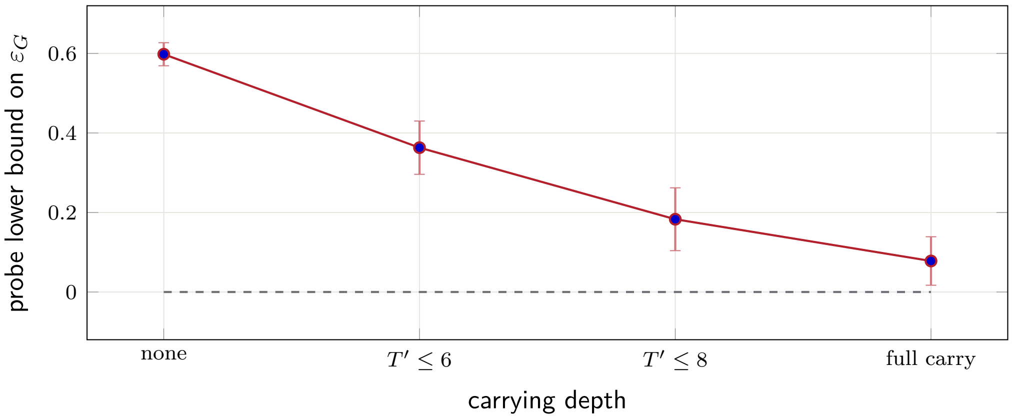

Figure 1: The dose-response of belief carrying. Each point is a certified lower bound on ε_G from the probe family (mirrored seeds, Wilson 95% intervals): 0.598 with no carrying (the stored profile, Paper II), 0.363 resolving only boundaries at $T' \leq 6$ (0.97 s/resolve), 0.183 at $T' \leq 8$ (3.3 s/resolve), and 0.078 $[0.016, +0.139]$ with full carrying (5.8 resolves/game). The full-carry point is measured at $n = 1000$; the shallower rungs at $n = 300$–400, where the Wilson half-width (≈ 0.11) is wider than several of the gaps between them.

class prior λ_κ, plays one round from the resolved strategy, and repeats. Successor leaves keep the stored values; the kernel, the class DAG, and the row structure are untouched.

Correctness checks. A prior-weight resolve reproduces the stored solution, so the change degenerates correctly; shadow replay reproduces the resolver's tables byte-exactly from the public record; full games run with zero fallbacks. Probing a deterministic resolver requires the probe's opponent model to be the resolver itself: the resolver-aware probe replays a deterministic shadow copy, checked to byte-exactness, so both sides of every comparison hold equally exact models.

Two findings. First, belief carrying gains nothing head-to-head: the resolver beats the stored profile 50.3% $[47.2, 53.4]$. An equilibrium of the correctly-believed subgame does not harvest a fixed opponent; only best response does. Second, carrying is a large *defensive* win (Figure 1). The cost anti-correlates with the benefit: the deep layers where TV is worst (0.92–0.95) are precisely the cheapest to resolve (≤ 1 s), which is what makes full carrying practical.

Why this certifies nothing. Three obstructions, each fatal alone.

1. Unsafe re-solving has no theorem. Each resolve derives μ_h assuming the agent's own past strategy; an opponent who knows this can deviate early to poison the assumption and cash in later. Known constructions in the re-solving literature make naive re-solving strictly *more* exploitable. Our probe found no such attack, which says more about the probe than about the agent.

2. The filter's opponent transfer is a model (Section 3); a guarantee must hold against opponents no filter models.

3. The leaves are G'-flavored. Every resolve prices its exits with stored values computed under class priors at deeper boundaries; their error *as G-values* is the dominant unknown,

and Section 7 measures it as substantial (0.146 worst-case at $T'=3$).

The structural inequality $\varepsilon_G \leq \varepsilon_{\text{comp}} + 2U \cdot \mathbb{E}[\text{TV}] \cdot \text{depth}$, with $\varepsilon_{\text{comp}}$ the profile's composed within-abstraction exploitability and depth the number of carried boundaries, is vacuous at the measured TV = 0.842 (right side ≈10 against a trivial cap of 2) and does not touch obstruction 1 at all. A certificate therefore requires a different agent, the one the theorem of Section 6 is about.

5 Safe re-solving

Fix the agent seat p and opponent q. The agent plays the stored root solve at the game root and, at every boundary it reaches, the strategy extracted from the gadget solve below. The *leaf functional* $\hat{v}_b(b', s', t') \in [-1, 1]$ prices round-end leaves (currently the stored values; Section 7 upgrades them), and terminal leaves are exact.

Definition 5.1 (Safe-resolving gadget)**.** At boundary b with own range g_b and any strictly positive weighting ν_b on the opponent's types, the gadget game $\tilde{G}(b)$ is: chance draws (s, t) with weight $K_b(s, t)\, g_b(s)\, \nu_b(t)$; the opponent observes t and chooses OUT, ending the game at the *promise* $w_b(s, t)$, or IN, after which the one-round subgame of $\kappa(b)$ is played with leaves priced by $\hat{v}_b$. The agent never opts out.

Define the conditional promise $\hat{w}_b(t) := \sum_s K_b(s, t)\, g_b(s)\, w_b(s, t) \,/\, M_b(t)$ with $M_b(t) := \sum_s K_b(s, t)\, g_b(s)$, and for any agent round strategy σ and opponent behavior τ the opponent's conditional counterfactual value $\tilde{V}_q^\tau(b, t; \sigma)$ in the gadget. The weighting ν_b cancels from every conditional quantity; it tunes convergence only.

Lemma 5.2 (Margin identity)**.** *Let σ_b be the agent's average strategy from the gadget solve and $m_b(t) := \max_\tau \tilde{V}_q^\tau(b, t; \sigma_b) - \hat{w}_b(t)$ its measured per-type margin (one exact best-response sweep, recorded live). Then $m_b(t) \geq 0$, and every opponent behavior in the gadget satisfies $\tilde{V}_q^\tau(b, t; \sigma_b) \leq \hat{w}_b(t) + m_b(t)$.*

Proof. The best response includes OUT, whose conditional value is exactly $\hat{w}_b(t)$, so the maximum is at least $\hat{w}_b(t)$; the inequality is the definition of the maximum. The sweep is exact: the opponent's in-round information sets are (public node, private state) pairs, swept by per-state argmax against path-summed agent reach, then a per-type maximum over pre-stage choices including OUT. □

Lemma 5.3 (Carry-down consistency)**.** *Construct promises and prices from one functional $\hat{v}$: at every resolve, $w_b := \hat{v}$ on the entry grid of $\kappa(b)$, and leaves priced by $\hat{v}_b$. If b' is a successor boundary reached from the resolve at b, with $g_{b'}$ the played-strategy transfer of g_b, then the conditional promise of the next resolve equals the conditional leaf price the b-resolve charged: $\hat{w}_{b'}(t') = \sum_{s'} K_{b'} g_{b'} \hat{v}_b(b', s', t') / M_{b'}(t')$.*

Proof. Both sides are the same $g_{b'}$-weighted average of the same matrix: the filter propagates its own factor under exactly the played strategy (the byte-exact shadow-replay check), and the opponent factor cancels from the conditional by Lemma 3.1. Verified as a float-precision identity on played traces. □

Remark 5.4 (One-sidedness, and why it matters). Lemma 5.3 is an equality for the one-functional construction, but the telescoping argument below survives promises *at or below* the parent's leaf conditionals: a promise may be tightened downward without breaking soundness. This freedom is what the “hybrid” promise experiments of Section 8 price, and their measured failure mode (tightening without enforcement is harmful) is one of the sharpest lessons of that section.

In implementation, a prior-weight gadget solve reproduces the stored value (margins under 20× solver tolerance), the compiled kernel and the reference implementation agree on the gadget game to 10^{-9} at identical iteration counts, safe shadow replay is byte-exact (10^{-12}), and full games run with zero fallbacks, including forced filter-break survival. The gadget’s conservatism costs 1.4 percentage points head-to-head against the unsafe resolver (confidence intervals overlapping) at $\varepsilon_{\max} = 8.9 \times 10^{-4}$ per resolve.

6 The composition theorem

Assumption 6.1. Full carry: every non-terminal boundary the agent reaches is resolved. A depth-limited agent adds, per unresolved round, an uncertified belief-reset term; the certificate is a full-carry object.

Assumption 6.2. No unrepaired filter break. Safe mode enforces this by construction (Section 3); the repair preserves every conditional the certificate consumes and cannot itself break.

Reachable chains are sequences $(b_1, t_1) \to \cdots \to (b_L, t_L)$ of successor boundaries and opponent types under the agent’s public dynamics; by Section 2 a game has at most $L_{\max} = 8$ resolved boundaries ($T' = 9, \ldots, 2$).

Theorem 6.3 (Composition, measured form)**.** *Under Assumptions 6.1 and 6.2, for every opponent strategy τ:*

$$u_q(A, \tau) \;\le\; B_q^{\text{root}} \;+\; \max_{(b_1,t_1)\to\cdots\to(b_L,t_L)} \sum_{i=1}^{L} m_{b_i}(t_i),$$

where B_q^{root} is the opponent’s exact best-response value against the stored root tables in the $\hat{v}$-priced root round, and the maximum ranges over reachable chains. For the mirrored agent pair, $\varepsilon_{\text{pair}} \le \varepsilon_{\text{round}}(\text{root}) + \sum_{\text{seats}} \max_{\text{chains}} \sum_i m_{b_i}(t_i)$.

Proof. Bottom-up induction on boundaries with $D(b,t) := m_b(t) + \max_{(b',t')\ \text{reachable}} D(b',t')$ (empty max $= 0$); the claim is $\text{CBV}_A(b,t) \le \hat{w}_b(t) + D(b,t)$.

Base: a resolve whose leaves are all terminal has $\hat{v}$ exact, so the gadget *is* the truth below b restricted to IN, and Lemma 5.2 closes it.

Step: fix τ below b. The true conditional value decomposes over the round’s leaf aggregates: writing ρ_τ for opponent-and-chance reach into the aggregate landing on (b', t'),

$$V_q^\tau(b,t) = \tilde{V}_q^\tau(b,t;\sigma_b) + \sum_{(b',t')} \rho_\tau \Big[V_q^\tau(b',t') - \tfrac{\sum_{s'} K_{b'} g_{b'} \hat{v}_b(b',s',t')}{M_{b'}(t')} \Big],$$

because gadget and truth differ only in leaf pricing, and the agent’s reach into each leaf under σ_b is exactly what defines $g_{b'}$. By Lemma 5.3 the subtracted term is $\hat{w}_{b'}(t')$; by induction the bracket is at most $D(b',t')$; by Lemma 5.2 the first term is at most $\hat{w}_b(t) + m_b(t)$. Take suprema.

Root: the same decomposition applied once to the stored root round, prior-weighted over root types, plus the zero-sum meter identity $B_0^{\text{root}} + B_1^{\text{root}} = \varepsilon_{\text{round}}(\text{root})$. □

Remark 6.4. The bound is opponent-model-free: the weighting ν_b and the filtered opponent factor influence which equilibrium the resolve finds, hence how small the margins come out, but never the validity of the theorem. This is exactly the property unsafe re-solving lacks.

End-to-end verification: the composed two-level agent (a gadget resolve at a T'=3 boundary plus per-history gadget resolves at its T'=2 successors) is transcribed into the flattened perfect-recall oracle and the opponent is exact-best-responded per type; the result matches $\hat{w}$ plus the chain-sum of recorded margins in 5/5 configurations (both seats, prior and tilted entries, including the worst promise-dishonor class), and carry-down consistency holds to 3×10^{-16}.

7 Value-of-belief functions and the exact frontier

The theorem charges margins, and margins are small only where promises are *honorable* (backed by something true in G). The stored values are not: at T'=3, 37 of 66 class values sit outside their exact-in-G brackets, worst case 0.146 (stored -0.657 against exact -0.510 $[-0.5105, -0.5100]$), reach-weighted 1.36×10^{-2}. This is the within-subgame belief-reset composition error, and it agrees with the full-carry probe residual (0.078), a cross-instrument consistency. The certificate therefore needs a new object: exact values of class subgames *as functions of entry belief.*

The domain correction. $V(\kappa, \mu)$ is not a function on the full belief simplex and is not piecewise-linear there. By Paper I's factorization (Thm. 10.3) the reachable domain is the rank-one family, and restricted to it V is piecewise-*bilinear* in the tilt pair (g_0, g_1): concave piecewise-linear in each tilt with the other fixed. A facet census confirmed the model: across sampled seat segments the worst midpoint-above-chord excess is negative (every midpoint at least 1.4×10^{-3} below chord plus certified slack).

The facet census. The operational complexity measure is the size of a greedy *strategy-library cover*: solved round strategies such that exact best-response sandwiches (lower and upper envelopes over library entries) close to 10^{-3} over a set of candidate beliefs. Full population at $T' \leq 3$:

	$T' = 2$	$T' = 3$
classes (canonical, pending-discard)	18	66
library size min / median / max	1 / 29 / 40	1 / 28.5 / 47
residual cover gap (max)	8.1×10^{-4}	9.9×10^{-4}
holdout gap median / max	2.3×10^{-3} / 0.196	4.5×10^{-3} / 0.657
stored value outside exact bracket	0/18	37/66
stored-minus-exact delta, median / max	1.2×10^{-4} / 2.0×10^{-4}	4.3×10^{-4} / 0.146

Facet counts are tens per class against the 10^5 threshold at which the approach would have been abandoned: the exact frontier is cheap where it was feared expensive. T'=2 stored values are exact in G (18/18 inside brackets), confirming the layer where $G' = G$. Covers are dense where mass goes and weak at belief-space vertices (holdout max 0.657), a weakness that returns with force in Section 8. At T'=4, buildability and reach *anti-correlate*: the most-visited classes exceed flat-tree build budgets, so deep-layer certification leans on functional pricing at T'=5 boundaries rather than exact T'=4 leaves for the heavy classes. Frontier validation is oracle-, mirror- and T'=2-containment-based; by the measurement above, stored-value equality is not required above T'=2.

Table 1: Per-layer certificate terms under the one-functional carry-down (171 resolves over 32 full-carry games): mean-of-max per-type margin plus mean resolve residual ε_b. Promises are exact in G at $T'{=}2$ and the margin term vanishes there; the stored surface at $T' \geq 4$ carries the distance.

Layer	margin (mean-of-max / max)	ε_b (mean)	price surface
$T'{=}9$	0.134 / 0.363	9.3×10^{-4}	stored
$T'{=}8$	0.133 / 0.734	9.1×10^{-4}	stored
$T'{=}7$	0.111 / 0.508	8.7×10^{-4}	stored
$T'{=}6$	0.106 / 0.310	2.1×10^{-4}	stored
$T'{=}5$	0.148 / 0.900	2.4×10^{-4}	stored
$T'{=}4$	**0.258** / 0.790	9.1×10^{-5}	stored (libraries pending)
$T'{=}3$	0.085 / 0.499	9.6×10^{-5}	frontier (prior point)
$T'{=}2$	**0.0003** / 0.0006	1.7×10^{-4}	frontier (*honorable*)

What the frontier owes the certificate. The theorem quantifies over all opponents, including ones steering to boundaries no probe visits, so the certificate needs an a-priori bound on $m_b(t)$ valid at every reachable (b, g_b). For each covered class κ and every member g of its factored entry family, three objects are required: an *honorable promise vector* $w_\kappa(g, \cdot)$, a certified guarantee that some playable library strategy caps the opponent's true conditional counterfactual value below the boundary at $w_\kappa(g,t) + r_\kappa(g,t)$ for all t simultaneously, with certified residual r; the *functional sandwich gap* $\Delta_\kappa(g,t)$ between the library's envelopes evaluated at the resolve's own entry ranges (two exact best-response sweeps per entry, $\sim$5 ms), the leaf-error norm of record; and an *off-path enforcement rule*, under which the resolver compares its solve's measured margins against the capping entry's and plays the better, converting measured margins into the a-priori form $m_b(t) \leq \varepsilon_b + r_{\kappa(b)}(g_b, t)$.

Corollary 7.1 (Certified form)**.** *With all three objects in place,*

$$\varepsilon_{\text{pair}} \;\leq\; \varepsilon_{\text{round}}(\text{root}) + \sum_{\text{seats}} \max_{\text{chains}} \sum_{i=1}^{L \leq L_{\max}} \left[\, \varepsilon_{b_i} + r_{\kappa(b_i)}(g_{b_i}, t_i) \,\right],$$

every term computed offline or recorded live.

8 Results: the certificate today

Everything above exists in code and is verified; this section reports what happens when it is all assembled. The assembly is where an earlier draft of this program fooled itself, and the audit that caught the error is part of the result.

One functional, measured margins. With the one-functional carry-down (a single price surface, frontier prior-entry values where covered and stored values elsewhere, used both as leaf pricing and as promises), Lemma 5.3 holds by construction at every layer and the measurement isolates exactly one quantity: honorability of the price surface.

Table 1 carries the whole story. Where the surface is honorable ($T'{=}2$: terminal-exact frontier values) the margin term collapses to gap scale, so the certificate mechanism works. Where it is the stored surface ($T' \geq 4$) each resolve charges 0.11–0.26, and layers 4–9 carry 91% of the per-seat chain. Resolve residuals (10^{-4} scale) are irrelevant everywhere; there is nothing to buy by solving harder.

Remark 8.1 (The leaf charge is intrinsic, 2026-08). The 0.11–0.26 per-resolve charge of Table 1 was later shown to be a property of depth-limited resolving from *any* stored belief-reset surface rather than a colour artifact of the incumbent store `solve-2-kernel`: pricing the resolves from a colour-exact store instead (`solve-4-qre-reach`, colour residual 3.5×10^{-9} against the incumbent's 1.67) left the probe's edge unchanged (0.122 against 0.078, $p = 0.32$, $n = 1{,}000$ per arm; the measurement of Remark 9.1). This is why the certificate pause stands even after the store of record changed in 2026-08 (Remark 8.4): a store the probe cannot beat still prices its resolves off a belief-reset surface, and the honorability obstruction is untouched.

Remark 8.2 (A rejected certificate). Summing Table 1's on-path terms over $L_{\max} = 8$ and both seats gives 1.958, and an earlier draft of this section (dated 2026-08-03 in the project records) reported that number as the certificate. An independent adversarial audit rejected it: the theorem's chain maximum ranges over *opponent-steerable* chains, and an on-path mean substitutes a self-play average for a supremum the opponent controls. The licensed assembly from the same data, the worst observed chain over visited boundaries only, is 8.22: vacuous. Reach-weighting is not a sound repair, because the opponent steers the carried beliefs and self-play samples a measure-zero slice of the chain space. The audit's verdict, adopted in full: certified ε_G today is the trivial cap. The sound averaging opportunities that survive (root-prior weighting over the opponent's root type, and a chance-expectation reproof of the successor step, whose 0.4–2.0 verification slack locates the loose inequality) are theorem work, listed in the program. The lesson: in certification, the assembly is as much a proof obligation as the per-piece inequalities, and an average is never a supremum.

Fixed promise surfaces cannot certify. The audit also produced the program's most decisive positive tool. Per-type dishonor of a *fixed* promise surface is linear-fractional in the own tilt, so its supremum over the rank-one family is attained at own-type vertices and can be certified exactly (a clairvoyant best response per vertex). The functional surface's certified supremum dishonor is median ≈1.0 and max 2.0 at *both* $T'{=}2$ and $T'{=}3$, even where on-path margins measure 3×10^{-4}. An earlier holdout measurement suggesting fixed refined surfaces might suffice (max 0.063) was a sampling artifact of insufficiently extreme tilts. Any certificate path through a fixed promise matrix is closed, permanently, at every layer. What remains open is *range-dependent* promises (library caps evaluated at the realized entry range) with per-type enforcement, whose residual is the library sandwich-gap supremum, itself lower-bounded by vertex probes at median 0.14–0.43 today; hence the vertex-completion item in the program.

Enforcement is mandatory. Honorable (tighter) promises push opponent opt-out mass up, and CFR's weighted objective then simply ignores opted-out types, whose margins are only weakly controlled by the solve. The measurement: hybrid promises (tightened by Remark 5.4) *worsen* the $T'{=}3$ margin term from 0.085 to 0.634 as built, because the meter measures against the tighter bar while the solve stops defending it. The audit's a-priori judgment ("sound but worthless without enforcement") was generous; the measurement says actively harmful. Post-solve per-type minimization with the capping entry (the enforcement rule of Section 7) is load-bearing machinery.

Remark 8.3 (Range-dependent leaf prices, and the fixed point they create, 2026-08). Since a fixed promise surface cannot certify, the leaf price must be evaluated at the successor's realized entry range. That range is a function of the agent's own play and the public record alone, never of the opponent's private choices, so it is admissible in a bound. Two requirements emerged in building it. The price must see the full public resolution, not merely the successor class: the bid chain and the location of the fatal flip both move the range while leaving the class alone. And the price must be a per-type CONSTANT rather than a per-pair matrix, because a parent contracts a leaf price against its own reaches, which are not the weights the cap was computed under; a first implementation with per-pair caps measured a *negative* excess (-5.4×10^{-3}), which is to say it did not cap. Per-type constants restored it ($+8.2\times10^{-5}$ and $+8.5\times10^{-5}$ in the two directions), which is the promise vector $\hat{w}$ of Lemma 5.2 arrived at from the other end.

With that in place the per-resolve residual factors exactly as $r = \varepsilon_b+$(successor sandwich at its realized range) verified to 10^{-5}. Measured at the class prior, with successors solved exactly at their realized ranges wherever a perfect-recall tree builds, the sandwich is 3.0×10^{-4} to 4.5×10^{-4} at T'=3 and 3.1×10^{-4} at T'=4, against the 6.3×10^{-3} per resolve Corollary 7.1 allows. Two rungs of the ladder are therefore flat and far inside the budget.

The extrapolation stops there, and an earlier draft of this remark did not. Exact successor solves exist only where a tree builds, which is $T'\leq 3$ entirely and T'=4 for 80 of 165 classes covering 34% of played mass. At T'=5 the successors must be priced from stored libraries, and the sandwich is 1.1×10^{-2}, thirty-five times the layers below. That figure confounds two causes: the ladder fattening, and the libraries being thin. It leans towards the latter, since enriching the library from four entries to all of them moves it from 3.2×10^{-2} to 1.1×10^{-2}, and since the hard T'=4 covers were themselves left open at residual 3.6×10^{-2} to 1.5×10^{-1} by an entry cap imposed for time. Layers 6 through 9, which carry most of the charge, remain untouched.

What binds is a fixed point: the price depends on the successor's range, the range on the parent's play, the play on the price. Solving afresh at each repricing oscillates (r swinging between 0.32 and 1.19). Warm-started repricing, damping, and dropping the running average each settle three of six measured classes at 1.5–2.2×10^{-4} and leave the other three unsettled; freezing the successor strategy stops the price moving but inflates the successor sandwich to 1.9, since a strategy committed at one range is worthless at another. Two natural culprits are ruled out by measurement: independently produced equilibria of the same successor at the same range move the per-type caps by at most 3.3×10^{-3}, and no leaf range is degenerate. Depth-limited solvers whose leaf values depend on the ranges the solve produces re-evaluate those values at the current iterate on every iteration; that variant is untried here.

A structural consequence outranks it. Pricing a leaf at T'=6 by recursion would require the T'=5 cap at its realized range, which requires a round sweep over T'=4 leaves priced at theirs, and so on; with 5.0×10^{4} public resolutions in a T'=9 round falling to 32 at T'=2, the product is astronomical. The certificate therefore needs a per-class, range-parameterized cap *function* at every layer, built once from the bottom and stored, so that pricing a leaf is a lookup. Storing the values is cheap, a few megabytes over all 2,860 classes. Storing the strategies the enforcement rule must commit to is not: a round strategy at T'=9 carries 2.2×10^{5} infoset rows. Constructing and pricing that ladder is the open problem, and the fixed point above governs how expensive each rung is rather than whether it exists.

The sandwich. Table 2 reports both jaws.

Statistical power, and a corrected number. The probe is a paired win-rate estimator, so its Wilson half-width is ≈ 0.11 at $n = 300$ and ≈ 0.06 at $n = 1000$ — in several places

Table 2: The certificate scoreboard. Lower bounds are probe results (mirrored seeds, Wilson 95%; $n = 300$ except where stated); the upper bound is the audited assembly. The target is 0.10. The last two probe rows are the 2026-08 numbers of record (Remark 8.4).

Instrument	value
probe LB, unsafe full carry (n=1000)	$\varepsilon_G \geq 0.078$ [0.016, 0.139]
probe LB, safe + stored promises	$\varepsilon_G \geq 0.160$ [0.047, 0.269]
probe LB, safe + certified $T' \leq 3$ promises	$\varepsilon_G \geq 0.113$ [0.000, 0.224]
probe LB, `solve-2-kernel`, guards on (n=3200)	$\varepsilon_G \geq 0.0306$ [-0.0040, $+0.0652$]
probe LB, `solve-6-cfrpolicy`, guards on (n=3200)	$\varepsilon_G \geq -0.0050$ [-0.0396, $+0.0296$]
certificate UB (audited)	**trivial** (2.0)

wider than the differences it is being asked about. An earlier revision of this paper reported the unsafe full-carry bound as 0.013 $[-0.099, +0.126]$ from $n = 300$ and described it as statistically indistinguishable from zero. Re-measured on the same seed family at $n = 1000$, the estimate is 0.078 [0.016, 0.139], whose interval excludes zero; two independent 500-game halves agreed to 0.004. The earlier figure was a low draw, not a different quantity, and the two intervals overlap. Readers should treat every row of Table 2 measured at $n \leq 400$ as carrying that half-width, and should not read differences smaller than it as real.

Two readings of Table 2 matter. The safe agent with dishonorable stored promises is *measurably more exploitable* (0.160) than the unsafe agent (0.078): safety machinery with bad promises is worse than no machinery, because the opponent can farm the dishonor. The probe converts exactly the margins of Table 1 into realized wins, confirming the dishonor is a real, farmable quantity rather than an accounting artifact. And certified promises at $T' \leq 3$ move the point estimate in the right direction ($0.160 \to 0.113$), but $n = 300$ cannot separate the variants; the certificate's upper bound is the instrument of record here, a deliberate allocation choice, since separating the variants at 95% would cost 10–20k games of compute that certifies nothing.

Remark 8.4 (Probe-zero endpoint, 2026-08). The lower jaw was re-measured in 2026-08 at $n =$ 3,200 games per arm on matched seeds, and two numbers supersede the 0.078 endpoint. First, the deployed incumbent's number of record is $\varepsilon_G \geq +0.0306$ $[-0.0040, +0.0652]$ (`solve-2-kernel` with play-time dominance guards on, full carry; artifacts `s5_probe_solve2_guards.json` and `_ext.json`); the 0.078 of Table 2's first row is the guards-off reading at $n = 1{,}000$, reproduced to four decimals inside the pooled guards-off figure of +0.0919 [+0.0573, +0.1262]. Second, the program's endpoint is a store this probe cannot beat: `solve-6-cfrpolicy`, which couples per-class re-solved CFR+ policies to the values of `solve-4-qre-reach` held fixed, reads $\varepsilon_G \geq -0.0050$ $[-0.0396, +0.0296]$ at the same n and seeds, a probe win rate indistinguishable from a coin flip; the artifacts are `probe_solve-6-cfrpolicy_a.json` and `_b.json`, pooled in `pool_solve-6-cfrpolicy_vs_solve2.json`. Its difference against the incumbent (-0.0356, $z = -1.43$, $p = 0.154$) is in the favorable direction and unresolved at this n, so the claim of record is non-inferiority, with the improvement extension pre-registered at 10–20k games per arm. The sandwich for the deployed agent therefore reads $[-0.005$ (the probe reads zero), 2.0 (trivial)$]$, with the lower jaw's CI as stated: the lower jaw is a statement about one probe family's reach, and nothing about the upper jaw has changed.

9 The program

Four items, each with the measurement that motivates it, in dependency order. The research risk is concentrated in item 2.

1. **Vertex-complete frontier libraries at $T' \leq 4$.** One exact solve per feasible own-type vertex (10–144 per class, sub-second each) collapses the library gap at vertices to solve tolerance; the interior supremum then follows from the piecewise-bilinear structure by small bilinear programs. Target: certified per-class residual r_κ at $T' \leq 4$ of order 5×10^{-3}. The vertex probes of Section 8 say the gap is real and say exactly where.

2. **Certified cap propagation upward.** A $T' \geq 5$ cap against a strategy is one kernel exact-best-response round sweep with capped leaves, sound by Theorem 6.3's machinery. The open question is whether six layers of propagation stay tight. Remark 8.3 reports two rungs flat and far inside budget, a third that is thirty-five times worse for reasons not yet separated, and the rest untouched. This remains the program's research risk, now with a cost model attached.

3. **Enforcement transcription.** Library entries transcribed to playable round strategies, with per-type minimization at resolve time. This converts measured margins into a-priori bounds and flips the hybrid experiment from harmful to a strict win.

4. **Sound accounting.** The averaging opportunities that survive the audit: margin deduplication (measured -0.007), a-invariant chain assembly (per-type margin vectors are persisted for this), root-prior weighting, and the chance-expectation reproof of the successor step.

Probe power becomes worth buying only after items 1–3 move the certificate; the next result lives in the upper jaw. Serving latency, multiplayer, public-statistic enrichment (ruled out by Paper II's refinement ladder), and further equilibrium-set selection (real-game noise, by Paper II's measurements) are out of scope for this program.

Remark 9.1 (Paused, with a lower bound only, 2026-08). Two measurements taken after this program paused settle how it stands. First, the hypothesis that the leaf-price dishonorability of Section 8 was largely the incumbent store's colour inconsistency in disguise was refuted: against a re-solve whose values are colour-exact to 3.5×10^{-9} where the incumbent is off by 1.67 (Paper II's third-generation solve), the resolver-aware probe's edge did not fall (0.122 $[0.060, 0.183]$ against the incumbent's 0.078 $[0.016, 0.139]$, $p = 0.32$, two-proportion test at $n = 1000$ per arm; `probe_resolver.py`, artifacts `probe_v3.json` and `probe_v10.json`). The dishonorability is therefore intrinsic to pricing depth-limited resolves from a stored belief-reset surface, and a better store cannot remove it. Second, the exact frontier surface that would replace those prices is infeasible above $T' \approx 4$ (Section 8). The statement of record follows: heads-up Red/Black ships with the measured lower bound $\varepsilon_G \geq 0.078$ $[0.016, 0.139]$ under full belief carrying (≥ 0.598 for the stored profile) and with no certified upper bound below the trivial cap of 2. This position is recorded deliberately, so that no reader infers an upper bound from the machinery's existence. The program resumes only if a new tool changes the cost of honest deep leaf prices; re-running it on a better store is now known to be pointless. As of 2026-08 the deployed store of record is `solve-6-cfrpolicy`, whose probe reading is zero (-0.0050 $[-0.0396, +0.0296]$ at $n = 3{,}200$; Remark 8.4), and the position stands unchanged: a lower bound only, with no certified upper bound below the trivial cap.

10 Conclusion

This paper proved a composition theorem for safe re-solving in heads-up Red/Black and built and verified the machinery the theorem charges: an exact belief filter, a gadget resolver with live-measured margins, and exact value-of-belief frontiers at the shallow layers. Measurement located the stored profile's entire observable exploitability in the belief reset: carrying beliefs drives the strongest probe's lower bound from 0.598 to 0.078 [0.016, 0.139]. The 2026-08 numbers of record extend that line to its endpoint: the guards-on incumbent reads +0.0306 [−0.0040, +0.0652] and the deployed store of record, `solve-6-cfrpolicy`, reads −0.0050 [−0.0396, +0.0296] at $n = 3{,}200$, a probe reading of zero (Remark 8.4). The certified upper bound today is the trivial 2, and an earlier claim of 1.958 was rejected by audit for substituting an on-path average for an opponent-controlled supremum. The distance between the jaws is localized in a single quantity, the honorability of the leaf-price surface at layers $T' \geq 4$. A four-item program, each item motivated by a measurement in this paper, targets exactly that quantity.

A Verification and reproducibility

House rule, unchanged from Papers I–II: no number enters this paper without a script that regenerates it. `v6_verify_composition.py` re-derives $L_{\max}$, checks Lemma 5.3 as a float-precision identity on played traces, and verifies Theorem 6.3 end-to-end against the perfect-recall oracle on composed two-level agents (5/5 configurations). The filter, gadget, shadow-replay, filter-break and kernel-parity checks run in the simulator's test suite; the facet census, fattening, margin-by-layer, vertex-supremum and probe measurements each write a JSON artifact whose values appear here verbatim. The perfect-recall oracle merges procurement flip interleavings (order irrelevance, Paper I), so the verified agent variant feeds its filter order-canonicalized records; this changes nothing in the theorem, which never references flip order.